ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B4 · EARTH AND SPACE

# Days, years and planets

★☆☆
Grade 5

**Learning goals**

- Recall facts about the Earth, Sun and Moon.
- Order the planets.

**1** **Fill in the gaps.** [6]

WORD BANK

**24** **12** **reflects** **axis** **west** **365¼** **east** **north**

The Earth spins on its axis once every 24 hours, causing day and night. It orbits the Sun once every 365¼ days. The Moon reflects light from the Sun. The Sun rises in the east and sets in the west.

**2** **Number the planets in order from the Sun (1 = closest).** [4]

4 Mars

7 Uranus

1 Mercury

2 Venus

6 Saturn

3 Earth

5 Jupiter

8 Neptune

## 3 True or false? [5]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. The Sun moves around the Earth every day. | | ✔ | The Earth spins; the Sun only appears to move. |
| 2. The Moon makes its own light. | | ✔ | It reflects sunlight. |
| 3. When it is summer in Bangladesh it is winter in Australia. | ✔ | | |
| 4. Shadows are shortest at midday. | ✔ | | |
| 5. The Sun is a star. | ✔ | | |

## 4 Match each word to its meaning. [4]

Draw a line to match each pair, or write the letter.

| | | |
|---|---|---|
| rotation ● | → A | ● **A** one spin of the Earth on its axis |
| orbit ● | → C | ● **B** a year with 366 days |
| leap year ● | → B | ● **C** the path of the Earth around the Sun |
| constellation ● | → D | ● **D** a pattern of stars |

ANSWER KEY

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LEARN · PLAY · WONDER

THINK IT THROUGH · B4 · EARTH AND SPACE

# Explaining what we see

★★☆
Grade 5

**Learning goals**

- Explain day, night and seasons with diagrams.
- Use models.

**1** **Shade the night side of the Earth and label day and night.** [2]

Sun on the left, Earth on the right

The half of the Earth facing away from the Sun is shaded and labelled night; the half facing the Sun is labelled day.

**2** **Explain why it is daytime in Dhaka when it is night-time in New York.** [2]

They are on opposite sides of the Earth: when Dhaka faces the Sun, New York faces away from it.

**3** **Explain why Bangladesh is hotter in June than in December. Use the words *tilted*, *direct* and *longer*.** [3]

In June the northern half of the Earth is tilted towards the Sun, so sunlight is more direct and the days are longer, giving more heat.

**4** **A model: you shine a torch (Sun) at a ball (Earth) with a stick through it (axis) and turn the ball.** [3]

**a)** What does turning the ball show?

day and night (rotation)

**b)** Give **one** way this model is not like the real Earth and Sun.

Sizes and distances are wrong; the torch only lights one direction, but the Sun shines in all directions; the ball is not moving around the torch.

**5** **Why does the Moon seem to change shape?** [2]

As it orbits the Earth we see different amounts of its sunlit half.

ANSWER KEY

INVESTIGATE · B4 · EARTH AND SPACE

# How does day length change through the year?

★★☆ Grade 5

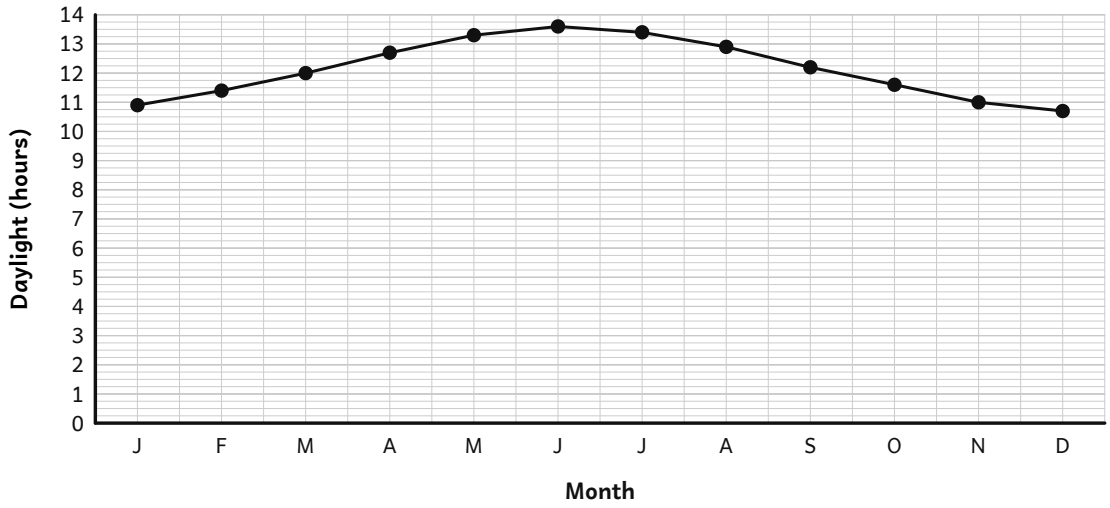
**Learning goals**

- Plot a line graph.
- Identify a pattern.
- Explain it using the Earth's tilt.

These are approximate hours of daylight in **Dhaka** in the middle of each month.

| Month | J | F | M | A | M | J | J | A | S | O | N | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Daylight (hours) | 10.9 | 11.4 | 12.0 | 12.7 | 13.3 | 13.6 | 13.4 | 12.9 | 12.2 | 11.6 | 11.0 | 10.7 |

**1** **Plot a line graph of daylight hours.** [4]




**2** **Answer using your graph.** [4]

**a)** Which month has the longest days? the shortest?

longest: June; shortest: December

**b)** Describe the pattern.

Days get longer from December to June, then shorter again: a pattern that repeats every year.

**c)** How much longer is a day in June than in December?

13.6 – 10.7 = 2.9 hours (almost 3 hours)

**3** **Predict the daylight hours in Singapore (on the equator) in June and in December. Explain.** [2]

About 12 hours in both months: near the equator day length hardly changes during the year.

**4** **Shadow stick. Plan how you would measure the shadow of a stick at different times of day. What would you keep the same?** [2]

Same stick, same place, measure at 9 am, 11 am, 1 pm, 3 pm with a metre stick; record length and direction; repeat on another sunny day.

ANSWER KEY

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REAL-WORLD CASE STUDY · B4 · EARTH AND SPACE

# Six seasons, one planet

★★★
Grade 5

**Learning goals**

- Compare seasons in different places.
- Link seasons to culture and farming.
- Explain using the Earth's tilt and position.

Science · Geography · Culture · Maths · Agriculture

Bengali tradition divides the year into **six seasons** (*ritu*): *Grishmo* (summer), *Borsha* (monsoon), *Shorot* (early autumn), *Hemonto* (late autumn), *Sheet* (winter) and *Boshonto* (spring). Farming follows the seasons: *Aman* rice grows in the monsoon and is harvested in Hemonto, celebrated with the **Nabanna** festival of new rice, while *Boro* rice grows in the dry winter with water from tube-wells.

Other places have very different seasons. In **Singapore**, near the equator, it is hot all year and every day has about 12 hours of daylight. In **Tromsø**, Norway, far to the north, the Sun does not set for about two months in summer and does not rise for about two months in winter. In **Sydney**, Australia, south of the equator, December is the middle of summer.

Day-length facts from standard astronomical tables (approximate).

**1** **Complete the table.** [6]

| Place | Where it is | What its seasons are like |
|---|---|---|
| Dhaka | north of the equator, near the Tropic of Cancer | hot, wet summer (monsoon); mild, dry winter; days 11–13.5 h |
| Singapore | on the equator | hot all year; about 12 h daylight every day |
| Tromsø | far north, near the Arctic | midnight sun in summer, polar night in winter |
| Sydney | south of the equator | summer in December, winter in June |

**2** **Explain why Sydney has summer in December, when Dhaka has winter.** [2]

In December the southern half of the Earth is tilted towards the Sun (summer there), while the northern half, with Dhaka, is tilted away (winter).

**3** **Choose one Bangladeshi festival or food linked to a season. Explain the link.** [2]

e.g. Nabanna: celebrates the new Aman rice harvest in Hemonto; pitha in winter made with new rice and date-palm molasses; mangoes in Grishmo.

**4** **Why does Boro rice need water from tube-wells, but Aman rice does not?** [2]

Boro grows in the dry winter with little rain; Aman grows in the monsoon when rain supplies the water.

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REVIEW AND CHALLENGE · B4 · EARTH AND SPACE

# Show what you know

★★★
Grade 5

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| R | S | U | N | W | P | Y | G | L | J |
|---|---|---|---|---|---|---|---|---|---|
| T | T | A | P | R | M | R | K | G | I |
| E | M | M | I | N | W | S | T | E | T |
| Q | Y | B | O | R | B | I | T | R | B |
| U | F | Y | C | O | H | I | B | W | J |
| A | R | G | F | W | N | O | F | C | H |
| T | I | L | T | T | M | S | T | A | R |
| O | J | L | P | L | A | N | E | T | Y |
| R | G | H | S | E | A | S | O | N | H |
| K | L | A | X | I | S | R | D | J | M |

**axis** **equator** **moon** **orbit** **planet** **season** **star** **sun** **tilt**

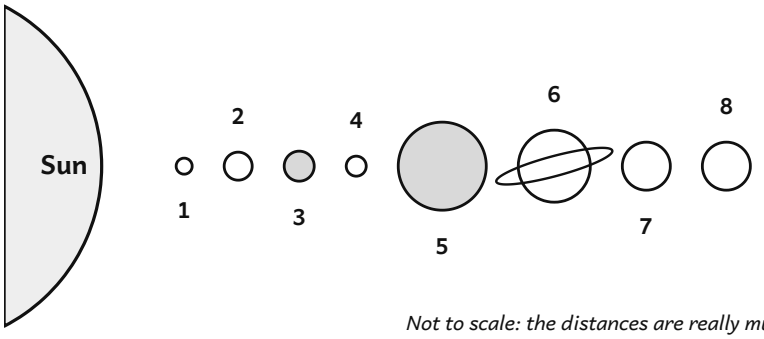
**2** **What causes day and night?** [1]

- ◯ the Earth orbiting the Sun
- ● **the Earth spinning on its axis**
- ◯ the Moon blocking the Sun
- ◯ clouds

**3** **What causes the seasons?** [1]

- ◯ the Earth's distance from the Moon
- ● **the tilt of the Earth's axis as it orbits the Sun**
- ◯ the Sun getting hotter
- ◯ the wind

**4** **Label the planets with numbers 1 to 8 from the Sun, then write the name of the largest planet.** [2]




*Not to scale: the distances are really much, much bigger.*

Jupiter

**5** **Challenge. Galileo saw moons going around Jupiter. Why was this important evidence?** [2]

It showed that not everything orbits the Earth, which supported the idea that the Earth and planets orbit the Sun.